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Global Constraints on Higgsing Localized Five-Dimensional Sectors in Compact M-Theory Vacua

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Abstract: Complex deformations of noncompact canonical Calabi–Yau threefold singularities encode Higgs-branch renormalization-group flows of five-dimensional theories. We determine which such flows remain available when several singularities occur in one compact Calabi–Yau threefold. The local sectors are conifold points supporting free hypermultiplets and cones over del Pezzo surfaces of degrees seven and six engineering the rank-one E_2 and E_3 fixed points. We identify their deformation coordinates with flavor moment maps and derive a compact Abelian gauging matrix from wrapped-M2 charges and divisor pairings. On either E_3 branch (the minimal nilpotent orbit closure of $\mathfrak{sl}_3$ or that of $\mathfrak{sl}_2$), we prove that the invariant deformation-coordinate values attained on the triplet moment-map zero locus in the rigid description are precisely the kernel of this matrix. The geometric comparison applies to Δ-regular anticanonical hypersurfaces whose fan polytopes have unit edges and only unimodular triangles, unit squares, or one-interior-point pentagons and hexagons as two-faces, with unit dual edges at the singular faces. Assuming trivial canonical bundle, $h^1(\mathcal{O}) = 0$, and the $\partial\bar{\partial}$-lemma on the crepant resolution, the same kernel is the image of the global first-order deformation space. Combining this identification with the mixed smoothing theorem gives a necessary and sufficient criterion: the kernel must meet the complement of every local discriminant. On the two-dimensional E_3 component this requires three separate nonzero coefficients, not merely a nonzero vector. The selected global deformation base is smooth when its relation space projects nontrivially to each selected E_3 summand; every tangent on that base integrates to a convergent deformation. Examples distinguish a joint conifold–E_2 transition, sectors forced to remain undeformed, and nonzero E_3 deformations forced to retain a singularity. These results concern geometric smoothing and invariant moment-map coordinates, not the full finite-Planck hypermultiplet geometry.

Keywords: M-Theory, Supersymmetric Gauge Theory, Supergravity Models, Differential and Algebraic Geometry

Contents

1 Introduction and summary

Topology-changing transitions in Calabi–Yau compactifications admit a particularly direct interpretation in M-theory. At a conifold degeneration, M2-branes wrapped on exceptional curves become massless. Their charges are the homology classes of those curves, and condensation along the Higgs branch connects the resolved and deformed Calabi–Yau threefolds [1–4]. Contractions of compact del Pezzo surfaces lead instead to interacting five-dimensional fixed points [5–8]. These two kinds of transition are usually analyzed separately: the former in terms of charged hypermultiplets and the latter in terms of the Coulomb and Higgs branches of a local superconformal field theory.

This paper studies compact transitions in which both types of sector occur at once. We call a transition *cooperative* when its global deformation space contains a direction with nonzero components in several local sectors but contains no nonzero direction supported on any one of those sectors alone. The basic question is global. Suppose that a compact Calabi–Yau threefold contains isolated singularities p which, in their noncompact geometric-engineering limits, define five-dimensional theories $\mathcal{T}_p$. Each local theory can possess one or more Higgs-branch directions. Which combinations of these directions remain available when the singularities are embedded in the same compact threefold and coupled to the vector fields of the compactification?

There is no reason for the answer to factorize. A complex deformation of the compact threefold restricts to a deformation of every singular germ, but the image of the restriction map

$$\mathbb{T}^1_X := \mathrm{Ext}^1_X(\Omega^1_X, \mathcal{O}_X) \longrightarrow \bigoplus_p T^1_{X,p}, \qquad T^1_{X,p} := \mathrm{Ext}^1_{\mathcal{O}_{X,p}}(\Omega^1_{X,p}, \mathcal{O}_{X,p}), \tag{1.1}$$

can be a proper subspace. The same phenomenon has a five-dimensional description. Reducing the M-theory three-form on harmonic two-forms gives Abelian vector fields A^I. Wrapped M2 states are electrically charged under these fields, while the compact embedding gauges Cartan subgroups of the flavor symmetries of the interacting sectors. The corresponding moment-map equations correlate local Higgs operators that would be independent in isolated noncompact models.

We make this relation precise for compact Calabi–Yau threefolds whose isolated singularities are ordinary double points and anticanonical cones over del Pezzo surfaces of degrees seven and six. We write

$$S_7 = \mathrm{Bl}_2\,\mathbb{P}^2, \qquad S_6 = \mathrm{Bl}_3\,\mathbb{P}^2. \tag{1.2}$$

The corresponding local sectors are a free hypermultiplet and the rank-one E_2 and E_3 fixed points, respectively. This notation avoids the common ambiguity between the degree of a del Pezzo surface and the number of blown-up points.

The local input is finer than a dimension count. In one complex structure, a nodal hypermultiplet with scalars $(h, \widetilde{h})$ projects to the conifold deformation coordinate $t = h\widetilde{h}$. For the E_2 theory, the invariant ring of its Higgs chiral ring under the complexified $SU(2)$ Cartan is

$$\mathbb{C}[\mathcal{H}_{E_2}]^{\mathbb{C}^*} \cong \frac{\mathbb{C}[J_0, U]}{(U^2, UJ_0)}. \tag{1.3}$$

This is the complete nonreduced miniversal base of the cone over S_7. The reduced smoothing coordinate is the $SU(2)$ Cartan moment map J_0; the additional nilpotent tangent direction is the $U(1)$ current-multiplet operator U. For E_3, taking invariants under the complexified maximal torus gives

$$\mathbb{C}[\mathcal{H}_{E_3}]^{T_{\mathbb{C}}} \cong \frac{\mathbb{C}[H_1, H_2, L]}{(H_1 L, H_2 L)}. \tag{1.4}$$

The plane (H_1, H_2) consists of the $SU(3)$ Cartan moment maps and the line L is the $SU(2)$ Cartan moment map. Equation (1.4) is the reducible miniversal base of the cone over S_6. Its plane is the maximal-torus invariant image $\mathcal{B}_{\mathfrak{sl}_3} \cong \mathbb{C}^2$ of the minimal nilpotent orbit closure $\mathcal{H}_{\mathfrak{sl}_3} := \overline{\mathcal{O}_{\min}(\mathfrak{sl}_3)}$; its line is the corresponding image $\mathcal{B}_{\mathfrak{sl}_2} \cong \mathbb{C}$ of $\mathcal{H}_{\mathfrak{sl}_2} := \overline{\mathcal{O}_{\min}(\mathfrak{sl}_2)}$. This description keeps the two full E_3 Higgs components and their invariant images separate rather than combining them into a single tangent-space count. The local rings follow from the known Higgs chiral rings [9]; magnetic quivers provide an independent component and Hilbert-series check [10–12].

Charge constraints and the local-to-global deformation map

Let $\widehat{X} \to X$ be a crepant resolution and let D_I be compact divisor classes. At a node, denote the exceptional curve by C_a. At an E_2 or E_3 point, let $R_{p\alpha}$ be a root in the summand of

$K_{S_p}^{\perp}$ selected by the chosen Higgs component. Here $K_{S_p}^{\perp}$ is the orthogonal complement of the canonical class in $H_2(S_p;\mathbb{Z})$ for the intersection pairing. The compact matrix is

$$c_{Ia} = D_I \cdot C_a, \qquad m_{Ip\alpha} = D_I \cdot \iota_{p*} R_{p\alpha}. \tag{1.5}$$

The first set of entries are the electric charges of wrapped M2-branes. The second set are the cocharacters specifying how A^I gauges the local flavor Cartans. Thus the complex moment maps take the form

$$\mu_I^{\mathbb{C}} = \sum_a c_{Ia} t_a + \sum_r m_{Ir}^{(7)} \beta_r + \sum_p \big(m_{Ip1} H_{p1} + m_{Ip2} H_{p2} + m_{Ip0} L_p\big), \tag{1.6}$$

with $H_{p1}L_p = H_{p2}L_p = 0$ at each E_3 point.

A *branch profile* σ chooses the full component $\mathcal{H}_{\mathfrak{sl}_2}$ or $\mathcal{H}_{\mathfrak{sl}_3}$ at each E_3 point; $\mathcal{B}_\sigma$ is the product of their invariant images with the nodal and reduced E_2 lines. Let $\mathcal{M}_{H,\sigma}^{\mathrm{rigid}}$ denote the product of the selected local Higgs cones, restricted to the triplet-moment-map zero locus for the effective compact Abelian group and then divided by that group. Restricting (1.6) to the invariant coordinates gives a linear complex map $Q_{X,\sigma}$. For the geometric equality below, X is a Δ-regular anticanonical hypersurface associated with an admissible four-dimensional reflexive polytope as defined in Section 4.2, and it satisfies (4.8). Our principal result is

$$\boxed{\mathrm{Im}\Big(\mathcal{M}_{H,\sigma}^{\mathrm{rigid}} \longrightarrow \mathcal{B}_\sigma\Big) = \ker(Q_{X,\sigma}) = \mathrm{Im}\left(\mathbb{T}_X^1 \longrightarrow \bigoplus_p T_{X,p}^1\right) \cap \mathcal{B}_\sigma.} \tag{1.7}$$

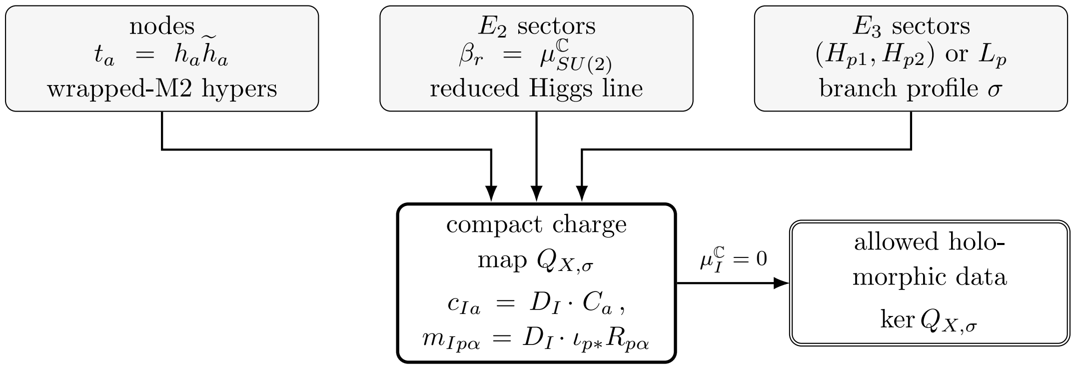
The first equality is a five-dimensional moment-map statement. The second is the geometric local-to-global deformation theorem. Equation (1.7) therefore identifies the geometric kernel with a physical charge and gauging constraint, rather than attaching a field-theory interpretation to a dimension computed after the fact.

Figure 1 summarizes the three local inputs and the common compact charge map.

The word "image" in (1.7) is essential. A local deformation base records invariant Higgs-branch coordinates, not the full hyperkähler cone or its metric. Likewise, the right-hand side is a complex linear space on a fixed reduced profile. We do not identify it with the full quaternionic-Kähler scalar manifold of the compact supergravity theory.

The real moment maps do not impose an additional inequality on this projected image. Every value of t_a, β_r, (H_{p1}, H_{p2}), or L_p has a balanced representative for which the corresponding local real Cartan moment maps vanish. Consequently every point of the complex kernel lifts to the zero-level triplet-moment-map locus. This also explains why signs in a homology relation do not obstruct Higgsing: phases reside in the elementary complex scalars, while their absolute values can be balanced.

The geometric comparison has an exact existence consequence. The mixed smoothing theorem of [13, Theorems 4.5 and 4.7], combined with (1.7), says that a convergent smoothing on a chosen profile exists precisely when the charge kernel contains a vector outside all local



The same divisor–curve pairing is the wrapped-M2 electric charge, the flavor-cocharacter gauging, and the dual of the geometric local-to-global map.

Figure 1. The compact Higgs-gluing mechanism on a fixed E_3 branch profile. The displayed kernel is the image in invariant complex moment-map coordinates. Balanced representatives lift it to the zero level of the full triplet moment maps. The exact geometric smoothing criterion requires this kernel to meet the complement of every local discriminant.

discriminants. At a node, an E_2 point or an E_3 line, the local coefficient must be nonzero. On an E_3 plane, in the root coordinates defined below, the condition is

$$H_1 H_2 (H_1 - H_2) \neq 0. \tag{1.8}$$

A nonzero plane vector alone is insufficient. Necessity holds for arbitrary analytic smoothing arcs, including those tangent to a discriminant to higher order. If the relation space projects nontrivially to every selected E_3 summand, the selected global deformation base is smooth and every tangent on it integrates. This hypothesis is automatic whenever the smoothing criterion holds; nodes and E_2 points require no additional projection hypothesis for smoothness of the selected base. These are geometric integrability statements, not a construction of the compact scalar metric.

Three compact phenomena

The theorem produces three qualitatively different effects.

First, a local Higgs direction can disappear completely after compact embedding. We give a compact model X_7 with one E_2 cone point. The reduced local theory has its $SU(2)$ Higgs component, but the compact kernel is zero. Hence no global first-order deformation activates it and no analytic smoothing exists.

Second, local sectors can Higgs only cooperatively. Our main explicit model X_9 contains two nodes and one E_2 point. Neither the two nodal sectors alone nor the E_2 sector alone has a nonzero global direction. In a primitive compact vector basis the effective charge matrix is

$$Q^{\text{eff}}_{X_9} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \ker Q^{\text{eff}}_{X_9} = \mathbb{C}(-1,-1,1). \tag{1.9}$$

The unique projected Higgs direction moves all three sectors together. An explicit family integrates it. The Hodge numbers change as

$$(h^{1,1}, h^{2,1}) = (6, 122) \longrightarrow (3, 123). \tag{1.10}$$

Two vectors are Higgsed by the rank-two gauging, while the third disappearing vector is the neutral rank-one E_2 Coulomb direction represented by the exceptional divisor. The surviving deformation supplies one neutral hypermultiplet, in agreement with (1.10).

Third, local smoothability need not imply simultaneous global Higgsability. A compact model X° contains 26 nodes, two E_2 points and two E_3 points. Every germ is locally smoothable. For all four ordered choices of $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ components at the two E_3 points, however, explicit primitive compact vectors force two nodal and both E_2 deformation coordinates to vanish. The projected kernels have dimensions $9, 10, 10, 12$, so many partial Higgs directions survive, but no profile contains a trajectory that activates all thirty sectors. The vanishing-period theorem promotes this conclusion beyond tangent space on every profile, because any smoothing must move the two forced nodes and both E_2 points whatever components it selects at the E_3 points. Here the superscript circle is part of the model name and does not denote the polar operation introduced in (4.7).

Relation to previous work and scope

Compact conifold charge matrices and their five-dimensional gauged-supergravity actions were developed in [4]. Malyshev's compact quintic supplied the closest geometric precursor: a del Pezzo root curve is homologous to four conifold curves, reducing the number of del Pezzo deformations [14]. Local E_n Higgs branches and magnetic quivers are treated in [9–12, 15]. De Marco, Del Zotto, Grimminger and Sangiovanni develop the local Higgs-branch stratification of five-dimensional fixed points from canonical singularities [16]. The present problem is the subsequent compact one: which invariant directions survive when several such sectors share the dynamical vectors of a single compact threefold. Compact gluing and multistage decoupling are analyzed in [17, 18].

Our result extends the mixed geometry in [14]. It gives a uniform theorem for free and interacting localized sectors and derives its matrix as the electric-charge and flavor-cocharacter gauging map. It also retains the two components of the E_3 Higgs cone, identifies the full scheme-level E_2/E_3 projections, and identifies the compact gauging with the relation map in the exact smoothing criterion. The geometric necessity, sufficiency and selected-base smoothness are results of [13]; the contribution here is their combination with the local operator dictionaries and dynamical compact gauging. The cooperative and obstructed compact examples test each part of this structure. None of these field-theoretic or branchwise statements is contained in the one-example deformation count of [14].

All compact harmonic modes used in the construction have finite norm at finite volume, so the full matrix in (1.6) is dynamical in the compact theory. In a limit that decouples gravity, only vector combinations with finite nonzero kinetic norm remain dynamical. The matrix is then projected to $Q_{\mathrm{dyn}} = P_{\mathrm{dyn}} Q_X$. Gravity may therefore decouple before all local sectors factorize; conversely, a fully factorized local limit removes the shared rows. We do not claim that the compact constraint survives every decoupling limit, and no example-specific Kähler scaling ray is needed for the finite-volume theorem.

The paper is organized as follows. Section 2 establishes the local operator dictionaries and magnetic-quiver checks. Section 3 derives the compact charge and gauging map. Section 4 states the first-order comparison, exact smoothing criterion and selected-base integrability theorem. Section 5 presents the absent, cooperative and obstructed Higgsing examples, including an obstruction that forces an E_3 deformation onto its discriminant. Section 6 separates the compact result from its possible rigid limits, and Section 7 discusses extensions and open problems.

2 Localized five-dimensional sectors and their Higgs coordinates

The compact argument requires the holomorphic flavor-moment-map coordinates that coincide with geometric deformation parameters. We identify these coordinates as affine schemes; the remaining Higgs-branch data of the rank-one theories will not enter the global charge map.

2.1 Geometric engineering and the meaning of the projection

M-theory on a noncompact canonical Calabi–Yau threefold singularity Y defines a five-dimensional theory $\mathcal{T}_Y$ [5, 7, 8]. A crepant resolution of Y parametrizes its Coulomb data, while normalizable complex deformations encode Higgs-branch data. This dictionary has been developed in many examples and can be refined to the stratification by symplectic leaves [12, 16].

Two qualifications will be important. First, the base $\mathrm{Def}(Y)$ of complex deformations is not itself the hyperkähler Higgs branch. In the examples below it is obtained from a chosen complex structure on the Higgs cone by taking affine invariants under a complexified flavor torus. It therefore records Cartan moment-map values but forgets the remaining orbit coordinates. Second, strict L^2 normalizability in an isolated noncompact metric is a separate analytic question. The compact theorem uses limits of modes on a compact threefold, where their norms are finite. We return to noncompact limits in Section 6.

The three local dictionaries are summarized in Table 1. Here $\mathcal{B}_{\mathrm{red}}$ denotes the reduced geometric deformation component selected by an ordinary complex Higgs vev.

2.2 The conifold sector

For the ordinary double point

$$Y_0 = \{xy - zw = 0\} \subset \mathbb{C}^4, \tag{2.1}$$

the small resolution replaces the singular point by a curve $C \cong \mathbb{P}^1$. An M2-brane wrapping C gives a five-dimensional hypermultiplet. In one complex structure let $(h, \widetilde{h})$ be its two

singularity	local 5d sector	$\mathcal{B}_{\mathrm{red}}$	invariant moment-map coordinates
ordinary double point	free hypermultiplet	$\mathbb{C}_t$	$t = h\widetilde{h}$
cone over $S_7 = \mathrm{Bl}_2\,\mathbb{P}^2$	rank-one E_2 theory	$\mathbb{C}_\beta$	$\beta = J_0$, the $SU(2)$ Cartan moment map
cone over $S_6 = \mathrm{Bl}_3\,\mathbb{P}^2$	rank-one E_3 theory	$\mathbb{C}^2_{H_1,H_2} \cup_0 \mathbb{C}_L$	(H_1, H_2) for $SU(3)$ and L for $SU(2)$

Table 1. The local holomorphic projection used in the compact theorem. The E_2 miniversal base also contains an additional nilpotent tangent direction generated by U; it is not an ordinary reduced Higgs trajectory.

complex scalars, with opposite charges under the relevant $U(1)$. The local complex moment map and the conifold deformation coordinate are the same invariant,

$$t = h\widetilde{h}, \qquad Y_t = \{xy - zw = t\}. \tag{2.2}$$

This local map can also be obtained from the type-IIA description of the rank-zero theory [19].

For later use, every $t \in \mathbb{C}$ has a representative with zero real moment map. If $t = |t|e^{\mathrm{i}\vartheta}$, take

$$h = \sqrt{|t|}e^{\mathrm{i}\vartheta}, \qquad \widetilde{h} = \sqrt{|t|}. \tag{2.3}$$

Then $h\widetilde{h} = t$ and $|h|^2 - |\widetilde{h}|^2 = 0$. This elementary observation is the local reason that arbitrary complex coefficients in a homology relation can solve the full triplet of moment-map equations.

2.3 The E_2 reduced component and nilpotent tangent

The cone over S_7 engineers the rank-one E_2 fixed point with flavor algebra

$$\mathfrak{e}_2 = \mathfrak{su}(2) \oplus \mathfrak{u}(1). \tag{2.4}$$

In a chosen complex structure, the Higgs chiral ring is generated by an $SU(2)$ moment-map triplet (J_+, J_0, J_-) and a further current-multiplet operator U, denoted P in [9]:

$$\mathbb{C}[\mathcal{H}_{E_2}] = \frac{\mathbb{C}[J_+, J_0, J_-, U]}{(J_0^2 - J_+J_-,\ U^2,\ UJ_+,\ UJ_0,\ UJ_-)}. \tag{2.5}$$

The reduction is the A_1 surface

$$\mathcal{H}_{E_2,\mathrm{red}} = \{J_0^2 = J_+J_-\} \cong \mathbb{C}^2/\mathbb{Z}_2, \tag{2.6}$$

and U defines an embedded point at its vertex.

The complexified $SU(2)$ Cartan has weights $(2, 0, -2, 0)$ on (J_+, J_0, J_-, U). A weight-zero monomial in the reduced ring is a power of J_0, because

$$J_+^a J_0^b J_-^a = (J_+J_-)^a J_0^b = J_0^{2a+b}. \tag{2.7}$$

Every nonzero monomial containing U is proportional to U. It follows that

$$\mathbb{C}[\mathcal{H}_{E_2}]^{\mathbb{C}^*} = \frac{\mathbb{C}[J_0, U]}{(U^2, UJ_0)}. \tag{2.8}$$

Altmann's miniversal base for the degree-seven cone is

$$\mathrm{Def}(C(S_7)) = \mathrm{Spec}\,\frac{\mathbb{C}[s_2, s_1]}{(s_1^2, s_1 s_2)} \tag{2.9}$$

[20]. The root marking from the crepant resolution fixes the dictionary, up to nonzero normalization,

$$\beta = s_2 \longleftrightarrow J_0, \qquad \alpha = s_1 \longleftrightarrow U. \tag{2.10}$$

Thus the complete nonreduced schemes agree, not only their tangent dimensions. The reduced line $\alpha = 0$ is the physical finite Higgs component. Since $U^2 = UJ_0 = 0$, the U direction exists over dual numbers but vanishes at every ordinary complex point; it cannot be counted as a second finite branch.

The Cartan interpretation is explicit in the double-cover coordinates:

$$J_+ = a^2, \qquad J_0 = ab, \qquad J_- = b^2, \qquad (a,b) \sim (-a,-b). \tag{2.11}$$

The Cartan acts as $(a,b) \mapsto (\lambda a, \lambda^{-1} b)$ and has moment maps

$$\mu^{\mathbb{C}}_{E_2} = ab = \beta, \qquad \mu^{\mathbb{R}}_{E_2} = \frac{1}{2}(|a|^2 - |b|^2). \tag{2.12}$$

The choice $a = \sqrt{|\beta|}e^{\mathrm{i}\arg\beta}$, $b = \sqrt{|\beta|}$ realizes any complex β while setting the real moment map to zero.

2.4 The two components of the E_3 sector

The cone over S_6 engineers the rank-one E_3 fixed point with flavor algebra

$$\mathfrak{e}_3 = \mathfrak{su}(3) \oplus \mathfrak{su}(2). \tag{2.13}$$

Its reduced Higgs branch is the union

$$\mathcal{H}_{E_3} = \underbrace{\overline{\mathcal{O}_{\min}(\mathfrak{sl}_3)}}_{\mathcal{H}_{\mathfrak{sl}_3}} \cup_{\{0\}} \underbrace{\overline{\mathcal{O}_{\min}(\mathfrak{sl}_2)}}_{\mathcal{H}_{\mathfrak{sl}_2}}. \tag{2.14}$$

The two components have quaternionic dimensions two and one. In terms of the complex flavor moment maps $\Phi \in \mathfrak{sl}_3$ and $\widetilde{T} \in \mathfrak{sl}_2$, the chiral-ring relations are

$$\Phi^2 = 0, \qquad \mathrm{tr}(\widetilde{T}^2) = 0, \qquad \Phi^i{}_j \widetilde{T}^{\dot\alpha\dot\beta} = 0 \tag{2.15}$$

[9]. The mixed relation says scheme-theoretically that positive-degree functions on different components annihilate one another. The intersection is one reduced point.

Let (H_1, H_2) be root-basis Cartan coordinates of Φ and let L be the Cartan coordinate of $\widetilde{T}$. The maximal-torus invariant rings of the two minimal nilpotent orbit closures are respectively $\mathbb{C}[H_1, H_2]$ and $\mathbb{C}[L]$. Taking invariants preserves the fiber product, because a complex torus is linearly reductive. Hence

$$\mathbb{C}[\mathcal{H}_{E_3}]^{T_\mathbb{C}} = \mathbb{C}[H_1, H_2] \underset{\mathbb{C}}{\times} \mathbb{C}[L] \cong \frac{\mathbb{C}[H_1, H_2, L]}{(H_1 L, H_2 L)}. \tag{2.16}$$

We reserve $\mathcal{H}_{\mathfrak{sl}_3}$ and $\mathcal{H}_{\mathfrak{sl}_2}$ for the full orbit closures and denote their invariant-coordinate images by

$$\mathcal{B}_{\mathfrak{sl}_3} := \operatorname{Spec} \mathbb{C}[H_1, H_2] \cong \mathbb{C}^2, \qquad \mathcal{B}_{\mathfrak{sl}_2} := \operatorname{Spec} \mathbb{C}[L] \cong \mathbb{C}. \tag{2.17}$$

The miniversal base of the degree-six cone is

$$\operatorname{Def}(C(S_6)) = \operatorname{Spec} \frac{\mathbb{C}[s_1, s_2, s_3]}{(s_1 s_3, s_2 s_3)}. \tag{2.18}$$

The compatible root marking gives

$$(s_1, s_2) \longleftrightarrow (H_1, H_2) \quad \text{on } \mathcal{B}_{\mathfrak{sl}_3}, \qquad s_3 \longleftrightarrow L \quad \text{on } \mathcal{B}_{\mathfrak{sl}_2}. \tag{2.19}$$

The mixed equations in (2.16) matter in the compact problem: at one E_3 point, an ordinary Higgs trajectory chooses the $SU(3)$ or $SU(2)$ moment maps, but not both.

With $K_{S_6}^{\perp}$ denoting the orthogonal complement of the canonical class in $H_2(S_6; \mathbb{Z})$ for the intersection pairing, the corresponding root decomposition inside the del Pezzo exceptional divisor is

$$K_{S_6}^{\perp} \cong A_2 \oplus A_1, \qquad R_1 = e_1 - e_2, \quad R_2 = e_2 - e_3, \quad R_0 = \ell - e_1 - e_2 - e_3. \tag{2.20}$$

Here (R_1, R_2) span A_2 and R_0 spans its orthogonal A_1. Root-basis coordinates are used throughout so that a charge functional with values (m_1, m_2) on the simple roots couples as $m_1 H_1 + m_2 H_2$, without an unwritten inverse Cartan matrix.

As in the E_2 case, arbitrary complex Cartan values admit representatives with zero real moment maps. One may use the rank-one Atiyah–Drinfeld–Hitchin–Manin (ADHM) presentation of a minimal nilpotent orbit closure: write $z_i = q_i \widetilde{q}_i$, impose $\sum_i z_i = 0$, and choose $|q_i| = |\widetilde{q}_i| = \sqrt{|z_i|}$ with phases reproducing z_i. This sets the ADHM and flavor real moment maps to zero pairwise. The $\mathfrak{sl}_2$ component is the same construction with two pairs.

2.5 Magnetic-quiver and Hasse-diagram check

The $\mathfrak{sl}_3$ and $\mathfrak{sl}_2$ components have the affine-A_2 triangle and affine-A_1 double-edge magnetic quivers, after removing the decoupled diagonal $U(1)$ [10–12]; the two-cone structure of the E_3 Higgs branch goes back to [5, 6]. Their Coulomb-branch dimensions are $3 - 1 = 2$ and $2 - 1 = 1$, agreeing with (2.19). With $x = t^2$, the Abelian monopole formula [21] gives

$$H_{\mathfrak{sl}_2}(x) = \frac{1 + x}{(1 - x)^2} = \sum_{n \geq 0} (2n + 1) x^n, \tag{2.21}$$

$$H_{\mathfrak{sl}_3}(x) = \frac{1 + 4x + x^2}{(1 - x)^4} = \sum_{n \geq 0} (n + 1)^3 x^n. \tag{2.22}$$

component	magnetic quiver	$\dim_{\mathbb{H}}$	coordinate	generic continuous stabilizer
$\mathcal{H}_{\mathfrak{sl}_3}$	affine A_2	2	(H_1, H_2)	$U(1) \times SU(2)$
$\mathcal{H}_{\mathfrak{sl}_2}$	affine A_1	1	L	$SU(3)$

Table 2. Independent local data for the two E_3 Higgs components. The $SU(2)$ (respectively $SU(3)$) factor complementary to the active $\mathfrak{sl}_3$ (respectively $\mathfrak{sl}_2$) orbit is unbroken. The table lists the full continuous generic stabilizer and omits only finite subgroups.

Since the components meet in a reduced point,

$$H_{E_3}(x) = H_{\mathfrak{sl}_3}(x) + H_{\mathfrak{sl}_2}(x) - 1. \tag{2.23}$$

The coefficient of x is $8+3 = 11$, the dimension of the $SU(3)\times SU(2)$ moment-map generators. This is independent of the compact deformation calculation.

The component data relevant below are collected in Table 2. Each minimal nilpotent orbit closure has only its open orbit and the origin as symplectic leaves. A generic point therefore has no transverse interacting endpoint; this local statement does not remove bulk vectors or other compact sectors.

3 Compact embedding and the five-dimensional gauging matrix

The physical matrix correlating the coordinates of Section 2 has two types of columns. Exceptional curves at nodes give electrically charged wrapped-M2 hypermultiplets. Roots in an exceptional del Pezzo surface determine how compact Abelian vectors embed into the flavor Cartan of the interacting fixed point. Both columns are obtained by evaluating curve classes on compact divisors.

3.1 Vector fields and wrapped-M2 charges

Let X be a compact singular Calabi–Yau threefold and let

$$\pi : \widehat{X} \longrightarrow X \tag{3.1}$$

be a projective crepant resolution. Choose harmonic two-forms ω_I on $\widehat{X}$ whose cohomology classes span the vector directions relevant to the transition. The M-theory three-form has the expansion

$$C_3 = \sum_I A^I \wedge \omega_I + \cdots . \tag{3.2}$$

If an M2-brane wraps a curve $C \subset \widehat{X}$, its worldline coupling is

$$\int_{\mathbb{R}\times C} C_3 = \sum_I \left(\int_C \omega_I\right) \int_{\mathbb{R}} A^I. \tag{3.3}$$

For a divisor class D_I dual to ω_I, the electric charge is therefore

$$c_I(C) = \int_C \omega_I = D_I \cdot C. \tag{3.4}$$

At a resolved node a, let $C_a \cong \mathbb{P}^1$ denote the exceptional curve. The wrapped M2-brane is the massless hypermultiplet of the conifold transition, with charge vector

$$c_{Ia} = D_I \cdot C_a. \tag{3.5}$$

Thus the map

$$\mathbb{Z}^{N_{\text{node}}} \longrightarrow H_2(\widehat{X};\mathbb{Z})/\text{tors}, \qquad e_a \longmapsto [C_a] \tag{3.6}$$

is simultaneously the homology map of the resolution and the electric charge map of the transition states. This is the standard compact conifold description [2, 4].

3.2 Flavor cocharacters from del Pezzo roots

Suppose the exceptional locus over p contains $S_{7,p}$ or $S_{6,p}$. The local non-Abelian flavor lattice is the root lattice inside $K_{S_p}^{\perp}$, the orthogonal complement of K_{S_p} in $H_2(S_p;\mathbb{Z})$ for the intersection form. Let

$$\iota_p : S_p \hookrightarrow \widehat{X} \tag{3.7}$$

be the inclusion and let $R_{p\alpha}$ be a root. The restriction $D_I|_{S_p}$ assigns degrees to the M2 states supported on curves in S_p. For E_2, with $R = e_1 - e_2$, the difference of the charges of the two exceptional-curve states is

$$c_I(e_1) - c_I(e_2) = D_I \cdot \iota_{p*}(e_1 - e_2). \tag{3.8}$$

This is the differential of the homomorphism from the compact $U(1)$ generated by A^I into the $SU(2)$ flavor torus. More generally, define

$$m_{Ip\alpha} = \left\langle D_I|_{S_p}, R_{p\alpha}^{\vee} \right\rangle \sim D_I \cdot \iota_{p*} R_{p\alpha}. \tag{3.9}$$

The proportionality allows for the conventional common normalization between a primitive geometric root and a Cartan generator. All algebras here are simply laced. We fix compatible root-marked moment-map coordinates throughout. Changing a root/coroot normalization or orientation changes the corresponding coordinate and matrix column together. The resulting kernels are identified by this invertible coordinate change; their coordinate vectors are not literally unchanged.

For E_2 there is one reduced root coordinate β_p. For E_3, the A_2 component uses (R_{p1}, R_{p2}) and the A_1 component uses R_{p0}, as in (2.20). Gauging a flavor isometry couples the vector-multiplet auxiliary triplet to the corresponding current-multiplet moment map [22]. Since moment maps add under diagonal gauging, the compact complex moment maps are

$$\mu_I^{\mathbb{C}} = \sum_a c_{Ia} t_a + \sum_r m_{Ir}^{(7)} \beta_r + \sum_p \left(m_{Ip1} H_{p1} + m_{Ip2} H_{p2} + m_{Ip0} L_p \right). \tag{3.10}$$

This conclusion uses only the flavor current multiplets of the interacting SCFT; it does not require a weakly coupled Lagrangian description of the E_2 or E_3 fixed point.

3.3 Branch profiles and the compact charge map

At each E_3 point choose either the full Higgs component $\mathcal{H}_{\mathfrak{sl}_2}$ or $\mathcal{H}_{\mathfrak{sl}_3}$. A collection of choices is a *branch profile* σ. The associated local coordinate space uses the invariant images of (2.17):

$$\mathcal{B}_\sigma = \bigoplus_a \mathbb{C}_{t_a} \oplus \bigoplus_r \mathbb{C}_{\beta_r} \oplus \bigoplus_p \mathcal{B}_{p,\sigma_p}, \qquad \mathcal{B}_{p,\mathfrak{sl}_2} = (\mathcal{B}_{\mathfrak{sl}_2})_p = \mathbb{C}_{L_p}, \quad \mathcal{B}_{p,\mathfrak{sl}_3} = (\mathcal{B}_{\mathfrak{sl}_3})_p = \mathbb{C}^2_{H_{p1},H_{p2}}. \tag{3.11}$$

Let Λ_{vec} be the lattice generated by the compact Abelian vector directions A^I, equivalently by their divisor classes D_I modulo the integral principal-divisor relations; the D_I are taken to span $H^2(\widehat{X};\mathbb{Q})$, which on the admissible class of Section 4.2 the restricted ambient toric divisors do. Its dual records one complex moment-map equation for each vector direction. The restriction of (3.10) to a profile is therefore the linear map

$$Q_{X,\sigma} : \mathcal{B}_\sigma \longrightarrow \Lambda^\vee_{\text{vec}} \otimes \mathbb{C}. \tag{3.12}$$

The same map can be defined before complexification. Let $\Lambda^{\text{H}}_{p,\sigma_p}$ be the root sublattice selected by the local component, and let a nodal summand be generated by $[C_a]$. Then

$$\begin{gathered} \Lambda^{\text{H}}_\sigma = \bigoplus_a \mathbb{Z}[C_a] \oplus \bigoplus_p \Lambda^{\text{H}}_{p,\sigma_p}, \\ Q^{\text{geo}}_{X,\sigma} : \Lambda^{\text{H}}_\sigma \longrightarrow H_2(\widehat{X};\mathbb{Z})/\text{tors}, \qquad \gamma \longmapsto \sum_p \iota_{p*}\gamma_p. \end{gathered} \tag{3.13}$$

Evaluating (3.13) on the divisor classes gives the matrix in (3.12). In other words, the geometric local-to-global map and the physical embedding of the local electric and flavor charges are dual descriptions of one lattice homomorphism.

In the toric hypersurface examples, the finite matrix is initially computed using ambient divisor classes. The geometric comparison reviewed in Section 4 proves that, under the admissibility hypotheses used here, its branch-restricted kernel equals the complete topological kernel. Thus the finite presentation does not miss a continuous constraint. Questions about a possible finite index between the printed toric lattice and the full integral cohomology lattice affect global forms and line operators, but not the rational kernel or the continuous Higgs selection rule.

3.4 The projected moment-map locus

Let G_{eff} be the compact Abelian group acting effectively on the transition sectors through (3.10). In the rigid zero-level description, form the triplet-moment-map quotient of the product of the local Higgs cones on a fixed profile. We use $\mathcal{M}^{\text{rigid}}_{H,\sigma}$ for this quotient and retain only its map to the invariant coordinates (3.11).

Proposition 3.1 (physical kernel)**.** *For every branch profile σ, the image of the rigid zero-level Higgs locus in its local deformation coordinates is*

$$\operatorname{Im}\Big(\mathcal{M}^{\text{rigid}}_{H,\sigma} \longrightarrow \mathcal{B}_\sigma\Big) = \ker(Q_{X,\sigma}). \tag{3.14}$$

Proof. If a point solves the complex moment-map equations, equation (3.10) states exactly that its invariant coordinate vector $b \in \mathcal{B}_\sigma$ obeys $Q_{X,\sigma} b = 0$. This proves one inclusion.

Conversely, take $b \in \ker Q_{X,\sigma}$. Lift every nodal coordinate t_a by (2.3). Lift every E_2 coordinate β_r by (2.11) with equal absolute values. On an E_3 component, use the balanced rank-one ADHM representatives described after (2.20). Every local real Cartan moment map then vanishes separately. The total real moment maps consequently vanish, while the total complex moment maps vanish because $Q_{X,\sigma} b = 0$. This constructs a point of the zero-level triplet-moment-map locus above every b in the complex kernel. Since the coordinates in $\mathcal{B}_\sigma$ are gauge invariant, taking the quotient does not change the image. □

If the selected local components have total quaternionic dimension d_{loc} and a rank-s Abelian group acts locally freely, the familiar count gives

$$\dim_{\mathbb{H}} \mathcal{M}_H = d_{\mathrm{loc}} - s = \dim_{\mathbb{C}} \ker Q_{X,\sigma}. \tag{3.15}$$

Equation (3.14) is stronger than this count: it specifies which combinations of coordinates survive and remains meaningful at profiles where the action has nontrivial stabilizers. It also avoids the incorrect identification of the full Higgs branch with a complex vector space.

3.5 Compact supergravity qualification

The microscopic charge assignment (3.4) and flavor embedding (3.9) are exact compactification data. Proposition 3.1 is stated in the rigid hyperkähler-cone description at zero moment-map level. The scalar manifold of five-dimensional $\mathcal{N} = 1$ supergravity is instead quaternionic Kähler, and its detailed metric receives gravitational and strong-coupling corrections. We use the rigid statement only to identify the holomorphic Higgs-deformation projection and its compact gauging constraint. We do not claim that it constructs the complete finite-Planck-mass scalar manifold.

This distinction does not turn the matrix into a background flavor constraint. At finite compact volume the fields A^I in (3.2) have finite kinetic terms and are dynamical. Their auxiliary-field equations impose the compact moment maps. What changes in a noncompact limit is the set of rows that remain dynamical, as discussed in Section 6.

4 Global deformation theorems

The physical matrix of Section 3 has an independent geometric characterization. We formulate it in terms of the topology of the singularity links and Milnor fibers. The mixed smoothing theorem then converts the charge-kernel calculation into an exact existence criterion. We distinguish this criterion from smoothness of the selected deformation base and from the nonlinear geometry of the physical Higgs branch.

4.1 Links, Milnor fibers and branch lattices

Let Σ_p be the link of the isolated singularity (X,p). For a smoothing component σ_p, let F_{p,σ_p} be a Milnor fiber and define the vanishing subspace

$$V_{p,\sigma_p} = \ker\big[H_2(\Sigma_p;\mathbb{Q}) \longrightarrow H_2(F_{p,\sigma_p};\mathbb{Q})\big]\,. \tag{4.1}$$

The spaces used in this paper are

$$\begin{array}{c|c|c} (X,p) & H_2(\Sigma_p;\mathbb{Q}) & V_{p,\sigma_p} \\ \hline \text{node} & \mathbb{Q} & \mathbb{Q} \\ C(S_7) & K_{S_7}^{\perp}\otimes\mathbb{Q} & A_1\otimes\mathbb{Q} \\ C(S_6),\ \mathcal{H}_{\mathfrak{sl}_2} & K_{S_6}^{\perp}\otimes\mathbb{Q} & A_1\otimes\mathbb{Q} \\ C(S_6),\ \mathcal{H}_{\mathfrak{sl}_3} & K_{S_6}^{\perp}\otimes\mathbb{Q} & A_2\otimes\mathbb{Q}. \end{array} \tag{4.2}$$

For a del Pezzo cone, the link is the unit-circle bundle of K_{S_p} and the Gysin sequence gives

$$H_2(\Sigma_p;\mathbb{Q}) \cong K_{S_p}^{\perp} \subset H_2(S_p;\mathbb{Q}), \tag{4.3}$$

where the superscript $\perp$ refers to the intersection pairing. The root subspaces in (4.2) are the cycles that die in the corresponding Milnor fiber (Proposition 2.1 and Section 2.4 of [23], where the three sweeping Fano threefolds are identified).

Choose a branch profile σ and set

$$K_\sigma(X) = \ker\left[\bigoplus_p H_2(\Sigma_p;\mathbb{Q}) \longrightarrow H_2(\widehat{X};\mathbb{Q})\right] \cap \bigoplus_p V_{p,\sigma_p}. \tag{4.4}$$

The map sends a nodal generator to its exceptional curve class and a del Pezzo link class to its pushforward from the del Pezzo exceptional divisor. Thus (4.4) is the rational kernel of (3.13).

Lemma 4.1 (local deformation–link comparison)**.** *Let $\widehat{X}_p \to (X,p)$ be the chosen crepant local resolution, with exceptional set E_p, and put*

$$K_p^{\mathrm{cr}} = \ker\Big[H^2_{E_p}(\widehat{X}_p;\Omega^2_{\widehat{X}_p}) \longrightarrow H^2(\widehat{X}_p;\Omega^2_{\widehat{X}_p})\Big]\,. \tag{4.5}$$

For a node and for the cones over S_7 and S_6, contraction with the crepant volume form and the boundary map give natural isomorphisms

$$T^1_{X,p} \cong K_p^{\mathrm{cr}} \cong H_2(\Sigma_p;\mathbb{C}). \tag{4.6}$$

These isomorphisms commute with the maps to $H^2(\widehat{X};\Omega^2_{\widehat{X}})$ and $H_2(\widehat{X};\mathbb{C})$ induced by the local inclusions. They carry the tangent cone of a smoothing component to $V_{p,\sigma_p}\otimes\mathbb{C}$.

For completeness, the branch assertion is finite. The circle-bundle Gysin sequence identifies the del Pezzo link with $K_{S_p}^{\perp}$. On the pentagon, the reflection fixing the unique smoothing parameter acts, after the determinant twist from contraction with the volume form, on the unique A_1 root line. On the hexagon, a sixfold rotation has a (-1) eigenspace and a Coxeter eigenspace; these are the A_1 line and A_2 plane. The local cohomology sequence identifies these eigenspaces with (4.5). This proves the branch statement and the compatibility with the Gysin maps; the full derivation is Lemmas 2.3 and 2.4 of [13], which use Schlessinger's comparison [24] in the depth-three form of [25].

4.2 A class of toric anticanonical hypersurfaces

Let $N \cong \mathbb{Z}^4$, let $M = \mathrm{Hom}(N, \mathbb{Z})$, and write $\langle\ ,\ \rangle$ for the natural pairing. For a four-dimensional reflexive polytope $\Delta \subset N_{\mathbb{R}}$, set

$$\Delta^\circ = \{u \in M_{\mathbb{R}} : \langle u, v\rangle \geq -1 \text{ for every } v \in \Delta\}. \tag{4.7}$$

Let Σ_Δ be the fan of cones over the faces of Δ and $\mathbb{P}_\Delta = \mathrm{TV}(\Sigma_\Delta)$ its Gorenstein Fano toric fourfold. A hypersurface $X \subset \mathbb{P}_\Delta$ is Δ-regular if its Laurent equation and every face restriction are nondegenerate in the sense of Batyrev [26].

If $G \subset \Delta$ is a two-face, its dual face $G^* \subset \Delta^\circ$ is an edge. A Δ-regular anticanonical hypersurface meets the torus orbit associated with G in $\ell(G^*)$ points, where ℓ is lattice length; the transverse germ at each point is the affine cone over the toric surface defined by G. We call G *singular* when it is not a unimodular triangle. The unit square, the unit-edge pentagon with one interior lattice point, and the analogous hexagon give, respectively, a node and the cones over S_7 and S_6.

The explicit finite theorem is proved for the following controlled class. We call Δ admissible if:

1. every two-face is a unimodular triangle, a unit square, or a pentagon or hexagon with one interior lattice point;
2. every edge has lattice length one; and
3. the dual edge of every singular two-face has lattice length one.

For a Δ-regular anticanonical hypersurface, the singular locus is then finite and consists exactly of nodes and cones over S_7 or S_6. The ambient divisor matrix detects the full rational topological kernel in (4.4); equivalently, the correction term in Batyrev's formula for $h^{1,1}$ [26] vanishes on this class (Proposition 8.3 of [23]).

The abstract local-to-global statement below uses the following additional hypotheses:

$$\omega_X \cong \mathcal{O}_X, \qquad h^1(X, \mathcal{O}_X) = 0, \qquad \widehat{X} \text{ satisfies the } \partial\bar{\partial}\text{-lemma}. \tag{4.8}$$

The third holds automatically for the projective crepant resolution fixed in Section 3.1; all three hold in the toric hypersurface examples of Section 5.

4.3 The first-order image theorem

Let $\mathbb{T}^1_X$ denote the global first-order deformation space of X. We view $V_{p,\sigma_p} \otimes \mathbb{C}$ as a subspace of $T^1_{X,p}$ through (4.6).

Theorem 4.2 (first-order local-to-global image)**.** *Let X arise from an admissible polytope and satisfy (4.8). For every branch profile σ,*

$$\operatorname{Im}\left[\mathbb{T}^1_X \longrightarrow \bigoplus_p T^1_{X,p}\right] \cap \bigoplus_p \big(V_{p,\sigma_p} \otimes \mathbb{C}\big) = K_\sigma(X) \otimes \mathbb{C}. \tag{4.9}$$

Equivalently, in the divisor and root bases of Section 3, the right-hand side is $\ker Q_{X,\sigma}$.

Proof. For a normal Gorenstein threefold with finite non-local-complete-intersection (non-lci) locus, the cotangent complex and Kähler-differential deformation groups agree through degree two (Lemma 2.2 of [13]). The local-to-global Ext sequence therefore begins

$$0 \longrightarrow H^1(X;\Theta_X) \longrightarrow \mathbb{T}^1_X \xrightarrow{\text{loc}} \bigoplus_p T^1_{X,p} \xrightarrow{\text{ob}} H^2(X;\Theta_X). \tag{4.10}$$

We identify the kernel of ob on each branch.

Let $E = \operatorname{Exc}(\widehat{X} \to X)$ and $X_{\text{reg}} = \widehat{X} \setminus E$. Excision and the local-cohomology sequence give the exact segment

$$H^1(X_{\text{reg}};\Omega^2_{X_{\text{reg}}}) \longrightarrow \bigoplus_p H^2_{E_p}(\widehat{X}_p;\Omega^2_{\widehat{X}_p}) \longrightarrow H^2(\widehat{X};\Omega^2_{\widehat{X}}). \tag{4.11}$$

By Lemma 4.1, each branch space $V_{p,\sigma_p} \otimes \mathbb{C}$ is a subspace of K^{cr}_p in the middle term. Moreover, the middle map in (4.11) fits into the commutative square

$$\begin{array}{ccc} \bigoplus_p H_2(\Sigma_p;\mathbb{C}) & \longrightarrow & H_2(\widehat{X};\mathbb{C}) \\ \downarrow \wr & & \downarrow \text{PD} \\ \bigoplus_p K^{\text{cr}}_p & \longrightarrow & H^4(\widehat{X};\mathbb{C}), \end{array} \tag{4.12}$$

where the lower arrow is the last map of (4.11) followed by the de Rham map. Thus a link class in the topological kernel (4.4) has zero de Rham image. The $\partial\bar{\partial}$-lemma makes $H^2(\widehat{X};\Omega^2_{\widehat{X}}) \to H^4(\widehat{X};\mathbb{C})$ injective, so exactness of (4.11) lifts that class to $H^1(X_{\text{reg}};\Omega^2_{X_{\text{reg}}})$.

Contraction with a trivialization of ω_X identifies $H^1(X_{\text{reg}};\Omega^2_{X_{\text{reg}}})$ with $H^1(X_{\text{reg}};\Theta_{X_{\text{reg}}})$. Schlessinger's depth-three comparison [24], in the form of [25], identifies the latter with $\mathbb{T}^1_X$, and naturality identifies the resulting localization map with loc in (4.10). This proves that every element of $K_\sigma(X) \otimes \mathbb{C}$ lies in the left-hand side of (4.9). Conversely, any global deformation class maps to zero under ob; the same commutative square sends its link class to zero in $H_2(\widehat{X};\mathbb{C})$. Lemma 4.1 then identifies its branch restriction with an element of $K_\sigma(X)$. The two inclusions prove the equality. This argument is the kernel comparison of Theorem 2.5 of [13], printed here to make the maps and obstruction group explicit. $\square$

Combining Theorem 4.2 with Proposition 3.1 yields the central identification

$$\operatorname{Im}\Big(\mathcal{M}^{\mathrm{rigid}}_{H,\sigma} \to \mathcal{B}_\sigma\Big) = \operatorname{Im}\left(\mathbb{T}^1_X \to \bigoplus_p T^1_{X,p}\right) \cap \mathcal{B}_\sigma. \tag{4.13}$$

This is the sense in which the geometric kernel is physically derived as a five-dimensional charge and gauging constraint.

4.4 The local discriminants and the exact criterion

A nonzero local deformation need not smooth a cone point. The *local discriminant* is the locus in a selected deformation component where the nearby fiber remains singular. At a node, an E_2 point or an E_3 line it is the origin. At an E_3 plane write

$$\kappa_p = H_{p1}(e_1 - e_2) + H_{p2}(e_2 - e_3) = \mu_{p1}e_1 + \mu_{p2}e_2 + \mu_{p3}e_3. \tag{4.14}$$

Thus $(\mu_{p1}, \mu_{p2}, \mu_{p3}) = (H_{p1}, H_{p2} - H_{p1}, -H_{p2})$. The discriminant consists of the three lines $\mu_{pi} = 0$, equivalently

$$H_{p1}H_{p2}(H_{p1} - H_{p2}) = 0. \tag{4.15}$$

The three coefficients are the smoothing coordinates of the three nodes in a simultaneous partial resolution of the degree-six cone [13, Section 3]. They also agree with the diagonal invariants $z_i = q_i\widetilde{q}_i$ of the local moment-map presentation in Appendix A.

Let $\mathcal{D}_\sigma \subset \mathcal{B}_\sigma$ be the union of the inverse images of these local discriminants under projection to the individual summands. The following is the degree-six/seven specialization of the mixed smoothing theorem [13, Theorem 4.7].

Theorem 4.3 (exact smoothing criterion)**.** *Let X be a connected normal projective complex threefold with $\omega_X \cong \mathcal{O}_X$ and $h^1(X, \mathcal{O}_X) = 0$, whose singularities are nodes and exact analytic anticanonical cones over S_7 or S_6. Fix a smooth crepant resolution, small over the nodes and standard divisorial over the cones, and a reduced branch profile σ. Then X has a convergent one-parameter smoothing inducing σ if and only if*

$$(K_\sigma(X) \otimes \mathbb{C}) \setminus \mathcal{D}_\sigma \neq \varnothing. \tag{4.16}$$

Equivalently, the relation space contains a vector nonzero on every rank-one summand and with all three μ_{pi} nonzero at each E_3 plane. Necessity applies to arbitrary analytic smoothing arcs, without an assumption on their order of contact with the local discriminants.

For necessity, the simultaneous partial resolution replaces an E_3 plane by three nodes without changing the noncentral fibers of a smoothing arc. Nodal period necessity supplies three separate nonzero coefficients; their sum is zero in the del Pezzo marking. The rank-one period argument treats the other germs. A generic rational linear combination of the resulting global relations makes all required coefficients nonzero simultaneously. This strengthens the scalar-period criterion of [23]. Sufficiency uses the selected-base smoothness stated below:

lift a relation satisfying (4.16) to a global tangent and integrate it. Every local discriminant factor has nonzero derivative, so all nearby fibers are smooth. The analytic globalization and arbitrary-arc necessity are proved in [13, Section 4].

Remark 4.4 (finite linear test). Condition (4.16) is equivalent to requiring that none of the individual scalar functionals defining $\mathcal{D}_\sigma$ vanish identically on $K_\sigma(X)$. A rational vector space is not a finite union of proper linear subspaces. Nonzero projection to an A_2 plane alone is weaker: the image can be one of its three discriminant lines.

4.5 Selected-base integrability and its physical scope

Fix analytic local classifying maps from a semiuniversal deformation of X. Denote by $\mathrm{Def}^\sigma(X)$ the scheme-theoretic inverse image of the product of the selected reduced local branches. The following specialization of [13, Theorem 4.5] concerns this selected space, not the unrestricted deformation scheme.

Theorem 4.5 (smooth selected deformation base)**.** *Under the hypotheses of Theorem 4.3, suppose $K_\sigma(X) \to V_{p,\sigma_p}$ is nonzero at every E_3 point. Then $\mathrm{Def}^\sigma(X)$ is smooth of dimension*

$$h^1(X, \Theta_X) + \dim_{\mathbb{Q}} K_\sigma(X), \tag{4.17}$$

and every tangent to it integrates to a convergent curve on the selected branches. No additional nonzero-projection hypothesis is required at nodes or E_2 points.

The geometric proof replaces an E_2 cone by a cone over $\mathbb{P}^1 \times \mathbb{P}^1$ and the E_3 line or plane by two or three nodes. The partial model has an unobstructed deformation space; analytic blow-down and a surjective map on selected tangents give the asserted smooth base. This addresses the non-complete-intersection cone germs rather than assuming the nodal unobstructedness theorem applies to them unchanged.

For the admissible toric class, Proposition 3.1 and Theorem 4.3 give

$$X \text{ smooths on } \sigma \quad \Longleftrightarrow \quad \mathrm{Im}(\mathcal{M}^{\mathrm{rigid}}_{H,\sigma} \to \mathcal{B}_\sigma) \setminus \mathcal{D}_\sigma \neq \varnothing. \tag{4.18}$$

Thus the compact gauging and the local discriminants decide existence of a geometric transition. A smooth selected base may nevertheless contain no smooth fiber. Nor does its smoothness imply that the nonlinear local classifying map is a linear inclusion, that every prescribed local arc globalizes, or that the full hypermultiplet manifold is a vector space. The embedded E_2 tangent is not an additional finite trajectory, and selected profiles with zero E_3 projection are not covered by Theorem 4.5.

The geometric theorem also covers exact cones of degree five and cones over $\mathbb{P}^1 \times \mathbb{P}^1$: their conditions are five nonzero conic-pencil pairings and a nonzero ruling-difference coefficient, respectively [13, Section 1.2]. Extending the physical operator dictionary to those sectors is separate from the node/E_2/E_3 comparison proved here.

5 Compact examples: forbidden, mixed and obstructed deformations

Three compact configurations exhibit distinct consequences of the charge map: a local Higgs flow that disappears, a mixed transition that exists only through cooperation, and a large system with many partial directions but no simultaneous all-sector trajectory. These effects depend on the actual kernel, not only its dimension.

5.1 A single E_2 direction that disappears globally

The smallest example, denoted X_7, is a compact anticanonical hypersurface with one isolated cone over S_7 and no other singularities. Locally, the E_2 theory has the reduced Higgs coordinate β of Section 2.3. In the convention (4.7), it is a Δ_7-regular hypersurface in $\mathbb{P}_{\Delta_7}$, where

$$\begin{aligned}\Delta_7 = \mathrm{conv}\{&v_0 = (1,0,0,0), v_1 = (0,1,0,0), v_2 = (0,0,1,0), v_3 = (-6,-4,-1,0),\\ &v_4 = (0,0,0,1), v_5 = (-6,-4,0,-1), v_6 = (-3,-2,1,-1)\}.\end{aligned} \tag{5.1}$$

This reflexive polytope has fifteen unimodular triangular two-faces and one unit-edge pentagon $G_7 = \mathrm{conv}\{v_2, v_3, v_4, v_5, v_6\}$ with the unique interior lattice point $p_7 = (-3,-2,0,0)$; its dual edge has length one. The three remaining boundary lattice points of Δ_7 lie in facet interiors, so their divisors miss a generic hypersurface. We fix a maximal projective crepant partial (MPCP) subdivision whose restriction to G_7 is the five-triangle star at p_7. All other two-faces are already unimodular, so this is the only local resolution choice entering the root calculation. Writing D_i for the ray divisor of v_i, the reduced A_1 root pairs with the ambient divisor classes as

$$b_{7,\beta} = -D_2 - D_3 + D_4 + D_5. \tag{5.2}$$

The row is nonzero and the reduced local branch is one-dimensional. Hence the root class has nonzero image in $H_2(\widehat{X}_7;\mathbb{Q})$, and therefore

$$K_{\mathrm{red}}(X_7) = 0. \tag{5.3}$$

The corresponding compact vector gauges the $SU(2)$ Cartan with nonzero coefficient. Its complex moment-map equation sets $\beta = 0$.

Equation (5.3) has two consequences at different logical levels. Theorem 4.2 says that no global first-order complex deformation moves the reduced local E_2 coordinate. The all-orders Theorem 4.3 then excludes a one-parameter smoothing whose first nonzero local term appears at higher order. Thus a genuine local Higgs branch of the isolated fixed point is absent from the compact deformation space.

This example is intentionally elementary. A single local sector supplies nothing with which its nontrivial root class can cancel. It provides the clean negative control for the cooperative example below.

5.2 A mixed smoothing requiring simultaneous local deformations

The main positive example X_9 contains two nodes and one cone over S_7. Order the reduced invariant coordinates as

$$x = (u_1, u_2, \beta)^{\mathsf{T}}, \tag{5.4}$$

where u_1, u_2 are conifold moment maps and β is the E_2 Cartan moment map. No nonzero relation among the two nodal curve classes alone has a nonzero coefficient on either curve. The E_2 root likewise cannot move by itself, as the preceding example illustrates.

The underlying reflexive polytope is

$$\begin{aligned}\Delta_9 = \mathrm{conv}\{&v_0 = (1,0,0,0), v_1 = (0,1,0,0), v_2 = (-1,-1,0,0),\\ &v_3 = (0,0,1,0), v_4 = (0,0,0,1), v_5 = (-4,-2,-1,0),\\ &v_6 = (-4,-2,0,-1), v_7 = (3,1,1,1), v_8 = (2,1,1,1)\}.\end{aligned} \tag{5.5}$$

Its singular two-faces are the squares $\{v_2, v_3, v_6, v_7\}$ and $\{v_2, v_4, v_5, v_7\}$ and the pentagon $\{v_3, v_4, v_5, v_6, v_8\}$. Star subdivision at the pentagon's interior point $p = (-2,-1,0,0)$ gives $E = D_p \cong S_7$. The only other boundary lattice point, $(-1,0,0,0)$, lies in a facet interior, and its divisor misses a generic anticanonical hypersurface. We fix an MPCP completion using the square diagonals (v_6, v_7) and (v_2, v_4), in the order just listed. With the sign convention that the chosen diagonal has coefficient -1, the two nodal circuit rows on the ten rays $(v_0, \ldots, v_8, p)$ are

$$v_2 + v_3 - v_6 - v_7, \qquad -v_2 - v_4 + v_5 + v_7. \tag{5.6}$$

The remaining choices in the MPCP completion lie over smooth strata and do not alter these circuit rows or the star of p. Principal-divisor relations eliminate D_0, D_1, D_3, D_4, giving the divisor basis below.

After removing a toric divisor that does not meet the generic hypersurface, a saturated basis of the toric divisor quotient is

$$\mathcal{D} = (D_2, D_5, D_6, D_7, D_8, E), \tag{5.7}$$

where $E \cong S_7$ is the exceptional divisor. This is a complete basis of $H^2(\widehat{X}_9; \mathbb{Q})$. In this basis, the compact gauging matrix is

$$Q_{\mathcal{D}} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{5.8}$$

An integral unimodular change of the six compact vector directions isolates two effective vectors,

$$V_1 = -D_6, \qquad V_2 = D_5, \tag{5.9}$$

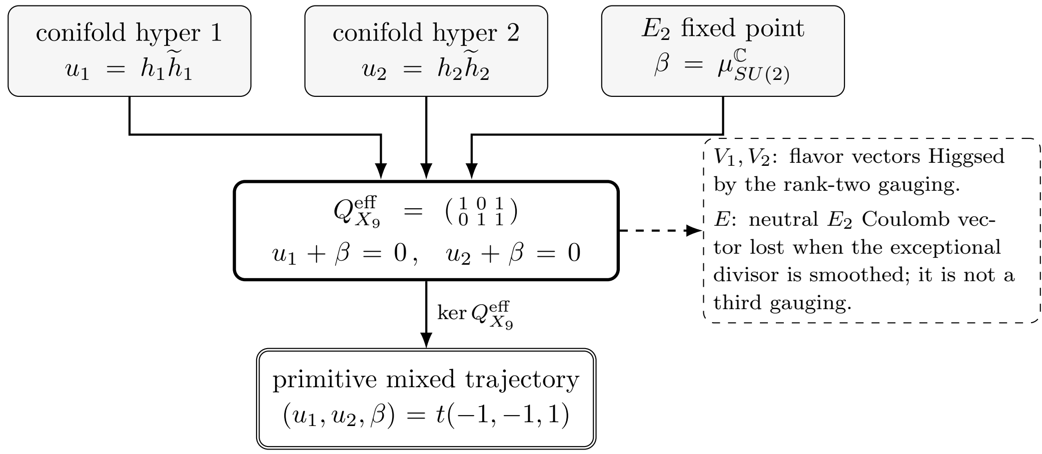


Figure 2. Cooperative Higgsing in X_9. No nonzero trajectory is supported on the two conifold sectors or on the E_2 sector alone. The unique global direction activates all three, while the protected vector-multiplet count separates the two Higgsed flavor vectors from the neutral SCFT Coulomb vector.

and brings (5.8) to

$$Q^{\text{eff}}_{X_9} = \begin{pmatrix}1&0&1\\0&1&1\end{pmatrix}. \tag{5.10}$$

Both nonzero Smith invariants are one in the printed toric lattice. The two complex moment-map equations are

$$u_1+\beta=0, \qquad u_2+\beta=0, \tag{5.11}$$

so

$$\ker Q^{\text{eff}}_{X_9} = \mathbb{C}(-1,-1,1). \tag{5.12}$$

The cooperative structure is displayed in Figure 2.

No nonzero vector in (5.12) is supported only on the nodes or only on the E_2 sector. The two free-hypermultiplet sectors and the interacting sector can move only together. Conversely, an explicit one-parameter Calabi–Yau family realizes the class

$$(u_1,u_2,\beta) = t(-1,-1,1) \tag{5.13}$$

and has smooth general fiber by Theorem 11.8 of [13]. More explicitly, take the facet $F = \{v_2, v_3, v_4, v_5, v_6, v_7, v_8\}$ and deformation degree $R = (0,1,-1,-1)$. On the ordered edges

$$(v_2v_5, v_2v_6, v_2v_7, v_3v_6, v_3v_7, v_3v_8, v_4v_5, v_4v_7, v_4v_8, v_5v_6, v_7v_8) \tag{5.14}$$

the edge-dilation vector is $\tau = (0,0,1,1,0,0,1,0,0,0,0)$. This is the compatible Minkowski decomposition used in the Ilten–Vollmert construction [27, 28]: it restricts to a segment-plus-segment decomposition on each square and to a segment-plus-unimodular-triangle decomposition on the pentagon. These data define the cited toric deformation and its relative anticanonical family. The kernel is therefore not only a first-order flat direction: it integrates to a mixed extremal transition, and it is cooperative in the sense defined in the Introduction.

There is a useful spectrum check. Exact lattice-point and Mayer–Vietoris calculations give

$$\begin{array}{c|ccc|cc} & h^{1,1} & h^{2,1} & \chi & n_V & n_H^{\text{neutral}} \\ \hline \widehat{X}_9 & 6 & 122 & -232 & 5 & 123 \\ X_{9,t} & 3 & 123 & -240 & 2 & 124. \end{array} \tag{5.15}$$

For a smooth compact Calabi–Yau threefold in M-theory, $n_V = h^{1,1} - 1$ and $n_H^{\text{neutral}} = h^{2,1} + 1$. The two effective gaugings in (5.10) Higgs two vector directions and remove two quaternionic directions from the product of local Higgs sectors. One quaternionic modulus remains, in agreement with the increase of n_H^{neutral} in (5.15).

The transition loses three, not two, massless vector multiplets. The third has a different origin. The exceptional-divisor vector E is neutral under the E_2 root moment map because

$$E|_E = K_E, \qquad \beta \in K_E^{\perp}. \tag{5.16}$$

It is the rank-one E_2 Coulomb direction and disappears when the entire surface is replaced by the smoothing. It is not a third flavor gauging. This separation between Higgsed vectors and the lost SCFT Coulomb vector is needed for the protected spectrum to agree with the topology.

The integrality statement above is made in the saturated toric divisor lattice. A possible finite index between that lattice and the full integral cohomology lattice would affect global forms or discrete remnants, but not the rank, the complex kernel or the multiplet count. We also do not claim the full massive spectrum at the interacting E_2 point.

5.3 Thirty locally smoothable sectors with no common trajectory

The large example X° contains

$$26 \text{ nodes}, \qquad 2\ E_2 \text{ points}, \qquad 2\ E_3 \text{ points}. \tag{5.17}$$

Every singular germ admits a local smoothing. At the ith E_2 point, let α_i denote the nilpotent scheme-theoretic tangent coordinate and β_i the reduced smoothing coordinate. At the jth E_3 point, let L_j be the A_1 coordinate and (H_{j1}, H_{j2}) the A_2 coordinates. Record a profile by the ordered pair of component labels selected at the two E_3 points. The four reduced profiles are

$$(2,2), \quad (2,3), \quad (3,2), \quad (3,3), \tag{5.18}$$

where 2 and 3 are subscripts naming the selected simple factor: 2 selects the $\mathfrak{sl}_2$ line and 3 the $\mathfrak{sl}_3$ plane.

Appendix B.2 gives the defining fan polytope Π, all singular two-faces, a strict convex height function fixing the crepant resolution, and the complete sparse charge matrix. Thus X° denotes a specified Π-regular anticanonical hypersurface rather than only an abstract singularity inventory.

profile	$\dim_{\mathbb{C}} \mathcal{B}_\sigma$	$\operatorname{rank} Q_\sigma$	$\dim_{\mathbb{C}} \ker Q_\sigma$	forced reduced coordinates
$(2,2)$	30	21	9	$u_9, u_{11}, \beta_1, \beta_2$
$(2,3)$	31	21	10	$u_9, u_{11}, \beta_1, \beta_2, L_1$
$(3,2)$	31	21	10	$u_9, u_{11}, \beta_1, \beta_2, L_2$
$(3,3)$	32	20	12	$u_9, u_{11}, \beta_1, \beta_2$

Table 3. The four branch-restricted compact charge maps for X°. In a profile (σ_1, σ_2), $\sigma_j = 2$ selects the $\mathfrak{sl}_2$ line and $\sigma_j = 3$ the $\mathfrak{sl}_3$ plane at the jth E_3 point; α_i is the nilpotent scheme-theoretic tangent coordinate and β_i the reduced smoothing coordinate at the ith E_2 point, while L_j is the $\mathfrak{sl}_2$ line coordinate at the jth E_3 point. Every nonzero Smith invariant is one in the printed divisor lattice.

In an integral divisor frame the full local-to-global matrix has size 36×26 and rank 21; the physical charge matrix is its transpose. Restricting to the four profiles gives the exact data in Table 3.

The forced-zero conclusions are witnessed by individual compact vectors, not only by a rank calculation. Five primitive integral vector combinations have complex moment maps

$$\begin{aligned} \mu^{\mathbb{C}}_{V_9} &= u_9 - \alpha_1 + \alpha_2, & \mu^{\mathbb{C}}_{V_{11}} &= u_{11} - \alpha_1 + \alpha_2, \\ \mu^{\mathbb{C}}_{V_{\beta_1}} &= \beta_1, & \mu^{\mathbb{C}}_{V_{\beta_2}} &= \alpha_2 + \beta_2, \\ \mu^{\mathbb{C}}_{V_{A_1}} &= L_1 + L_2. \end{aligned} \tag{5.19}$$

At an ordinary complex point of the E_2 Higgs scheme, $\alpha_1 = \alpha_2 = 0$. The first four equations in (5.19) therefore force

$$u_9 = u_{11} = \beta_1 = \beta_2 = 0 \tag{5.20}$$

on every reduced profile. The fifth equation correlates L_1 and L_2 on $(2,2)$, isolates the remaining line coordinate on $(2,3)$ or $(3,2)$, and becomes inactive on $(3,3)$. This explains the rank drop in the last row of Table 3.

The correct conclusion is not that X° has no Higgs directions. Its projected kernels have dimensions between nine and twelve. Rather, no profile contains a point at which all thirty local sectors have nonzero smoothing coordinates. The all-orders result is stronger than this first-order observation. Any one-parameter smoothing of X° would smooth the nodes N_9 and N_{11} and both E_2 points, whatever components it selects at the two E_3 points. Theorem 4.3 and Remark 4.4 then require a class in the kernel of the induced profile with nonzero projection to the coordinates in (5.20), and Table 3 excludes such a class on every profile. Consequently no analytic one-parameter family smooths all thirty germs, even if its local parameters begin at different orders. This is Theorem 6.5 of [23], whose Corollary 6.7 also excludes every formal arc meeting all thirty local smoothing loci.

5.4 Nonzero deformations constrained to a discriminant

There is a second obstruction mechanism: an allowed E_3 deformation can be nonzero but confined to a discriminant line. We use the compact example of [13, Theorems 5.1 and 5.3], with thirty nodes and two degree-six cone points. This is a different threefold from X°, which has twenty-six nodes and four cone points. The cited construction specifies its reflexive fan polytope, projective crepant resolution and full divisor–curve matrix. Here we record the three divisor identities that explain its obstruction as compact gauging.

Let $n_0, \ldots, n_{29}$ be its nodal coordinates. At cone $i = 0, 1$, write λ_i for the A_1 coordinate and (a_i, b_i) for the coefficients of $A_i = e_3 - e_1$ and $B_i = e_2 - e_3$ in its A_2 plane. The three plane coefficients are

$$a_i A_i + b_i B_i = -a_i e_1 + b_i e_2 + (a_i - b_i) e_3, \qquad a_i b_i (a_i - b_i) \neq 0 \tag{5.21}$$

for a smoothing direction. Three integral compact divisor combinations give the moment-map equations

$$\begin{aligned} n_{13} - a_0 + b_0 - a_1 + b_1 &= 0, \\ b_0 + b_1 &= 0, \\ n_{16} + \lambda_0 + \lambda_1 &= 0. \end{aligned} \tag{5.22}$$

They hold on the full charge kernel before a profile is chosen. Denote the line and plane by L and P. On LL the first equation forces $n_{13} = 0$, while on PP the third forces $n_{16} = 0$. On LP the second forces $b_1 = 0$; on PL it forces $b_0 = 0$. Every profile therefore fails the exact smoothing criterion.

The mixed profiles nevertheless contain relations nonzero at every singular point. For example the cited LP relation has $(\lambda_0, a_1, b_1) = (3, 1, 0)$ and all thirty nodal coefficients nonzero. Its E_3 plane deformation is nonzero, but its middle coefficient in (5.21) vanishes. Thus a nonzero invariant deformation at each sector does not imply simultaneous smoothing. The selected bases are smooth of dimensions $34, 34, 34, 35$ on LL, LP, PL, PP, respectively, and all their tangents integrate [13, Section 5.3]. The failure is to leave the local discriminants, not a failure of selected-base integrability. This does not exclude partial Higgs directions or identify the entire physical vacuum space with a deformation base.

5.5 Finite classification for polytopes with at most nine vertices

The three examples are part of a finite, reproducible classification. Among the 77 admissible reflexive four-polytopes with at most nine vertices in the Kreuzer–Skarke list [29] and a degree-seven cone point, the all-orders necessity condition obstructs 76; 67 of these have a single singular point, while nine have one node in addition to the E_2 point. The remaining case is X_9, whose explicit mixed family is smooth. Thus cooperative Higgsing is the unique smoothable outcome in this precisely stated bounded class by Proposition 9.2 and Corollary 6.7 of [23] and Theorems 11.8 and 10.3 of [13].

This census shows that the compact constraint is systematic rather than an accident of one matrix. It is not a statistical sample of all compact Calabi–Yau threefolds, and no universality claim is inferred from the ratio 76/77.

6 Normalizability and gravity-decoupling limits

The compact theorem and a rigid noncompact limit impose different normalizability conditions. In the compact model, the rows of Q_X arise from dynamical vector fields. In a noncompact limit, some of those fields can become background flavor sources, in which case their moment-map equations must be removed. We formulate the distinction in this section.

6.1 Finite compact volume

On a smooth compact resolution, the standard reduction (3.2) gives the five-dimensional kinetic term

$$S_5 \supset -\frac{1}{4}\int_{M_5} G_{IJ} F^I \wedge *_5 F^J, \qquad G_{IJ} \propto \int_{\widehat{X}} \omega_I \wedge *_6 \omega_J. \tag{6.1}$$

[30]. Every harmonic form on a compact manifold has finite norm. After separating the graviphoton, these fields supply the $h^{1,1}(\widehat{X})-1$ vector multiplets. At the transition point the graviphoton direction is the Kähler class of the singular threefold, which pairs to zero with every exceptional curve and root, so its row of Q_X vanishes and the effective rows in Section 5 are vector-multiplet directions. The pairings in (3.5) and (3.9) are therefore charges under dynamical fields, not charges under nondynamical flavor backgrounds.

Similarly, the kinetic metric for compact complex-structure moduli is schematically

$$G_{\kappa\bar{\lambda}} \propto \frac{\int_X \chi_\kappa \wedge \overline{\chi}_{\bar{\lambda}}}{\int_X \Omega \wedge \overline{\Omega}}, \qquad \chi_\kappa \in H^{2,1}(X). \tag{6.2}$$

[30]. The local coordinates in $\mathcal{B}_\sigma$ are limits of these compact directions, subject to the global image theorem. Thus no additional local L^2 assumption is needed to impose

$$Q_X b = 0 \tag{6.3}$$

at finite compact volume.

6.2 Dynamical and background vectors in a local limit

Consider a one-parameter family of Kähler classes $J(\Lambda)$ with

$$\mathrm{Vol}_{J(\Lambda)}(X) \longrightarrow \infty, \qquad G_{N,5} \longrightarrow 0. \tag{6.4}$$

For a vector combination $v = v^I D_I$, let $\omega_v = v^I \omega_I$ and define its kinetic norm

$$\mathcal{N}_v(\Lambda) = \int_X \omega_v \wedge *_{J(\Lambda)} \omega_v, \tag{6.5}$$

understood in a finite-normalization basis after the standard five-dimensional Einstein-frame rescaling. The vector belongs to the retained dynamical theory only when its normalized kinetic term has a finite nonzero limit. Geometrically, this is the requirement that it become an L^2 two-form on the retained noncompact space. A compact divisor may furnish such a form; a divisor which becomes noncompact gives a non-normalizable flavor background.

Let P_{dyn} project the compact vector lattice onto the combinations that remain dynamical, and let $\mathcal{B}_{\text{loc}}$ be the product of the local invariant-coordinate spaces retained in the chosen noncompact limit. The moment-map matrix of the limiting theory is

$$\boxed{Q_{\text{dyn}} = P_{\text{dyn}} Q_X, \qquad \operatorname{Im}(\mathcal{M}_H \to \mathcal{B}_{\text{loc}}) = \ker Q_{\text{dyn}}.} \tag{6.6}$$

If the norm of a shared vector diverges, its canonically normalized coupling to the local sector vanishes. That vector is no longer integrated over in the local theory; its moment-map equation is not imposed, and the corresponding row of Q_X is deleted. Equation (6.6) is therefore the limit of the compact derivation, not an independent ansatz.

6.3 Multistage decoupling

Several local singularities can remain connected after the overall Calabi–Yau volume is sent to infinity. Shared compact surfaces or lower-codimension singular loci can continue to gauge a common flavor subgroup. Only a further decompactification factorizes the individual local theories. This multistage structure appears in geometric constructions of 5d theories and in compact analyses of generalized symmetries [17, 18].

For the present mechanism there are accordingly three regimes:

regime	dynamical shared vectors	constraint
compact supergravity	all compact rows	$Q_X b = 0$
glued local QFT	selected finite-norm rows	$P_{\text{dyn}} Q_X b = 0$
factorized local sectors	no shared rows	no cross-sector equation.

(6.7)

Coupling to gravity is therefore sufficient but not necessary for the correlation. A gravity-decoupled glued QFT may retain it as an ordinary 5d gauging. What is special to the compact statement is that every row is dynamical before a limit is chosen.

6.4 Complex deformations and the local L^2 caveat

The same distinction applies to Higgs coordinates. Local geometric-engineering work uses a protected notion of dynamical complex deformation that is checked against Higgs chiral rings and magnetic quivers [12, 16]. Strict L^2 normalizability of a representative three-form can additionally depend on the asymptotic metric of the chosen noncompact Calabi–Yau, with subtleties already visible for the conifold. We therefore separate:

1. compact normalizability, which is sufficient for the theorem here;
2. protected local Higgs dynamics, established for the node, E_2 and E_3 components used above; and
3. strict local L^2 realization in a specified noncompact metric, which is not proved in this paper.

No Kähler scaling ray has yet been constructed for X_9 or X° which determines exactly which shared rows survive after gravity decouples. Accordingly, our principal theorem concerns compact M-theory at finite volume. Claims about a particular rigid limit must use (6.6) and supply that additional geometric analysis.

7 Discussion

The compact selection rule has a simple physical origin. Local Higgs sectors carry electric charges or flavor moment maps. A compact Calabi–Yau does not leave all of the corresponding flavor symmetries global: its C_3 vectors gauge diagonal combinations determined by divisor–curve pairings. Their moment-map equations cut out a correlated subspace of the product of local deformation coordinates. The same divisor–curve map appears in the local-to-global deformation sequence, so the five-dimensional vacuum equation and the geometric smoothing relation have the same kernel.

For nodes this reduces to the familiar conifold transition. The substantive extension is that an interacting del Pezzo sector enters the relation on the same footing as a wrapped-M2 hypermultiplet, through a current-multiplet moment map rather than an elementary field bilinear. The X_9 relation

$$(u_1, u_2, \beta) = t(-1, -1, 1) \tag{7.1}$$

is the simplest realization: two conifold condensates and an E_2 Higgs operator participate in one primitive compact transition. Neither part exists as a global transition without the other.

The E_3 sector shows why a branchwise formulation is necessary. Its local Higgs branch is not irreducible. The components $\mathcal{H}_{\mathfrak{sl}_3}$ and $\mathcal{H}_{\mathfrak{sl}_2}$ couple to different compact cocharacters, have different dimensions, and can change the rank of the global gauging matrix. Replacing the local theory by $\dim T^1 = 3$ would erase precisely the field-theoretic information responsible for the profile dependence of X°.

7.1 Relation to generalized-symmetry gluing

Compact M-theory models also correlate the generalized symmetries and global forms of several local sectors. There, Mayer–Vietoris maps determine which defects and higher-form symmetries glue consistently [18]. The analogy with the present result is structural. In both cases, independent local data are replaced by the kernel or cokernel of a map supplied by the complement of the singular neighborhoods in the compact threefold.

The constrained objects are different. Generalized-symmetry calculations track defect groups, polarizations and higher-group structures. Our matrix acts on ordinary Higgs moment maps and complex deformation coordinates. One result does not follow from the other. Together they indicate that the global topology of a compactification controls more than the spectrum of isolated local theories: it also controls how their moduli and symmetry data can be assembled.

7.2 Comparison with the compact del Pezzo–conifold precursor

Malyshev established that a root curve in a compact del Pezzo surface can be homologous to curves at spatially separated conifold points, and that this relation reduces the number of del Pezzo deformations [14]. That calculation proves the geometric phenomenon in a particular quintic. It does not derive the relation as a five-dimensional gauging or formulate a theorem for interacting Higgs branches.

The branchwise charge-kernel theorem is therefore a strict extension, both in scope and in physical content. It applies uniformly to the admissible toric class and identifies the relation matrix with electric and flavor gaugings. It retains the reduced and embedded E_2 schemes and both E_3 components, determines the exact image of the moment-map locus, and combines it with the mixed smoothing criterion of [13]. The resulting existence test uses the local discriminants as well as the charge kernel. The positive and negative examples include an independent protected-spectrum check. The antecedent is the mixed homology relation; the new result is its branchwise five-dimensional kernel theorem.

7.3 Open problems

The exact smoothing criterion and selected-base integrability resolve the geometric existence question under the stated hypotheses, including the two-dimensional E_3 component. They also distinguish abstract smoothings from a chosen ambient construction. For example, the threefold X_{19} of [13, Corollary 9.3], with fourteen nodes and one degree-seven cone point, has a smooth full deformation base of dimension thirty and admits a smoothing, although every single-degree ambient family in the specified class fails to smooth it. Failure of such an ambient construction is therefore not a compact gauging obstruction.

The remaining questions concern the following distinctions.

1. *Unrestricted deformation schemes.* Smoothness of a selected base does not determine how distinct profiles meet, the full nilpotent structure at E_2 points, or selected bases with zero E_3 projection. These are outside the smoothness assertion used here, not gaps in the exact criterion for existence of a smoothing.

2. *Controlled rigid limits.* For X_9, the shared flavor gauging is lost in every gravity-decoupling limit with bounded local Kähler scales [31, Section 4]. Other compact models, including X°, require their own kinetic and Kähler-cone analysis; the compact existence criterion alone does not decide which gaugings survive a noncompact limit.

3. *Integral global form.* The examples determine primitive Smith data in their toric divisor lattices. Establishing index one in full integral cohomology is separate from the rational smoothing criterion. It is established for X_9 in [31, Section 3 and Appendix B]; other models require an integral analysis before discrete remnants or line operators can be inferred.

4. *The compact scalar geometry.* Equation (4.13) identifies a protected holomorphic projection. Constructing the full coupled quaternionic-Kähler space and its chiral-ring analogue would go beyond the rigid zero-level description.

5. *Broader sectors.* The geometric theorem already extends beyond toric hypersurfaces to projective threefolds with its exact analytic germs, including degree-five and $\mathbb{P}^1 \times \mathbb{P}^1$ cones. Extending the local physical dictionaries to those sectors, and then to non-isolated singularities and higher-rank theories, requires additional operator and charge data.

The mixed-transition mechanism already follows from the results above. Higgs directions that are independent in isolated local theories need not remain independent in a compact M-theory vacuum. Dynamical five-dimensional vectors select their allowed combinations, and a global transition can require cooperation between free conifold states and an interacting SCFT sector. This is the compact generalization of the conifold charge relation.

A Local invariant rings and balanced representatives

This appendix records two local calculations used in Proposition 3.1.

A.1 E_3 maximal-torus invariants

The closure of the minimal nilpotent orbit of $\mathfrak{sl}_n$ has the rank-one presentation

$$\overline{\mathcal{O}_{\min}(\mathfrak{sl}_n)} = \operatorname{Spec}\left(\frac{\mathbb{C}[q_1, \ldots, q_n, \widetilde{q}_1, \ldots, \widetilde{q}_n]}{(\sum_i q_i \widetilde{q}_i)}\right)^{\mathbb{C}^*_{\mathrm{ADHM}}}, \tag{A.1}$$

where the ADHM torus acts with weights $+1$ and -1 on q_i and $\widetilde{q}_i$. The complex moment-map matrix is

$$\Phi^i{}_j = q_i \widetilde{q}_j, \qquad \operatorname{tr}\Phi = 0. \tag{A.2}$$

The flavor torus acts by $q_i \mapsto t_i q_i$ and $\widetilde{q}_i \mapsto t_i^{-1}\widetilde{q}_i$, with $\prod_i t_i = 1$. After combining it with the ADHM torus, an invariant monomial has equal powers of q_i and $\widetilde{q}_i$ for each i. The invariant ring is therefore generated by

$$z_i = q_i \widetilde{q}_i, \qquad \sum_i z_i = 0. \tag{A.3}$$

Since a complex torus is linearly reductive, taking invariants is exact and introduces no additional relation. Thus

$$\mathbb{C}[\overline{\mathcal{O}_{\min}(\mathfrak{sl}_3)}]^{T_{SU(3),\mathbb{C}}} \cong \frac{\mathbb{C}[z_1, z_2, z_3]}{(z_1 + z_2 + z_3)} \cong \mathbb{C}[H_1, H_2], \tag{A.4}$$

$$\mathbb{C}[\overline{\mathcal{O}_{\min}(\mathfrak{sl}_2)}]^{T_{SU(2),\mathbb{C}}} \cong \frac{\mathbb{C}[w_1, w_2]}{(w_1 + w_2)} \cong \mathbb{C}[L]. \tag{A.5}$$

One convenient root-basis convention is

$$H_1 = z_1, \qquad H_2 = -z_3, \qquad L = w_1 = -w_2. \tag{A.6}$$

The E_3 chiral ring is the fiber product of the two orbit rings over their origins. Its invariant ring is consequently

$$\mathbb{C}[H_1, H_2] \underset{\mathbb{C}}{\times} \mathbb{C}[L] \cong \frac{\mathbb{C}[H_1, H_2, L]}{(H_1 L, H_2 L)}. \tag{A.7}$$

A.2 Balanced real moment maps

For arbitrary $(H_1, H_2) \in \mathbb{C}^2$, solve (A.3) and (A.6) by

$$z_1 = H_1, \qquad z_2 = -H_1 + H_2, \qquad z_3 = -H_2. \tag{A.8}$$

For each $z_i = |z_i| e^{\mathrm{i}\theta_i}$ choose

$$q_i = \sqrt{|z_i|}\, e^{\mathrm{i}\theta_i}, \qquad \widetilde{q}_i = \sqrt{|z_i|}. \tag{A.9}$$

Then $q_i \widetilde{q}_i = z_i$ and $|q_i|^2 - |\widetilde{q}_i|^2 = 0$ for every i. The ADHM real moment map and all flavor-Cartan real moment maps vanish. The $SU(2)$ component is identical with $w_1 = L, w_2 = -L$. Equations (A.9), (2.3) and (2.12) supply the local balanced lifts used in the proof of Proposition 3.1.

A.3 Abelian monopole sums for the E_3 components

Removing the diagonal magnetic charge from the affine-A_1 quiver leaves $m \in \mathbb{Z}$. Its monopole sum is

$$\frac{1}{1-x} \sum_{m \in \mathbb{Z}} x^{|m|} = \frac{1+x}{(1-x)^2}. \tag{A.10}$$

For the affine-A_2 triangle, set one diagonal charge to zero. The sum is

$$\frac{1}{(1-x)^2} \sum_{m_1, m_2 \in \mathbb{Z}} x^{\frac{1}{2}(|m_1 - m_2| + |m_1| + |m_2|)} = \frac{1 + 4x + x^2}{(1-x)^4}. \tag{A.11}$$

The two series are respectively the character dimensions of the $SU(2)$ representations of highest weight $2n$ and the $SU(3)$ representations with Dynkin labels (n, n). Adding the two component rings and subtracting their common constant gives (2.23).

B Integral charge data for the compact examples

B.1 The X_9 divisor reduction

In the ordered basis (5.7), the three local coordinate rows are

$$B_{\mathrm{red}} = \begin{pmatrix} 1 & 0 & -1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 \end{pmatrix}, \qquad Q_{\mathcal{D}} = B_{\mathrm{red}}^{\mathsf{T}}. \tag{B.1}$$

The following integral vector combinations define a unimodular basis:

$$\begin{aligned} V_1 &= -D_6, & V_2 &= D_5, \\ S_1 &= D_2 + D_7, & S_2 &= D_2 + D_5 + D_6, \\ S_3 &= D_8, & \Phi &= E. \end{aligned} \tag{B.2}$$

The determinant of the change-of-basis matrix is -1, and in this frame

$$Q' = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{B.3}$$

This proves both the effective matrix and its primitivity in the printed toric lattice.

The full E_2 tangent scheme also contains the nilpotent coordinate α. Its row is the pairing of the divisors with a class of $K_{S_7}^{\perp}$ complementary to the root; the choice of complement is immaterial on the reduced branch, where $\alpha = 0$. In the divisor order (5.7), the row is

$$(0, -2, 1, 0, -2, 0). \tag{B.4}$$

Some spectator combinations can therefore couple to α over dual numbers. This does not add a finite Higgs mass because $\alpha^2 = \alpha\beta = 0$ and α vanishes on the reduction.

B.2 The X° toric and resolution input

Let $N_\circ \cong \mathbb{Z}^4$ be a fresh one-parameter-subgroup lattice and set $\Pi = \mathrm{conv}\{r_1, \dots, r_{26}\} \subset (N_\circ)_\mathbb{R}$. Thus Π is the fan polytope: the ambient fourfold is defined by the fan over its faces, and X° is a Π-regular anticanonical hypersurface. In particular, Π is not the polar $\Delta^\circ \subset M_\mathbb{R}$ of (4.7). The following table is the complete vertex list; its order fixes all subsequent labels.

(B.5)

1	$(-1,-1,-1,-1)$	14	$(0,-1,-1,0)$
2	$(-1,-1,-1,0)$	15	$(0,-1,0,1)$
3	$(-1,-1,0,-1)$	16	$(0,-1,0,0)$
4	$(-1,-1,0,1)$	17	$(0,1,-1,-1)$
5	$(-1,-1,1,0)$	18	$(0,1,-1,0)$
6	$(-1,-1,1,1)$	19	$(0,1,0,-1)$
7	$(-1,0,-1,-1)$	20	$(0,1,0,0)$
8	$(-1,0,-1,0)$	21	$(1,1,-1,-1)$
9	$(-1,0,0,-1)$	22	$(1,0,-1,0)$
10	$(-1,0,0,1)$	23	$(0,0,0,-1)$
11	$(-1,0,1,0)$	24	$(0,0,0,1)$
12	$(-1,0,1,1)$	25	$(0,0,1,0)$
13	$(0,-1,-1,-1)$	26	$(0,0,1,1)$

The twenty-six unit-square faces, written by vertex index, are

$$
\begin{array}{c|c c c|c}
N_1 & \{1,2,7,8\} & & N_{14} & \{5,6,25,26\} \\
N_2 & \{1,2,13,14\} & & N_{15} & \{7,8,17,18\} \\
N_3 & \{1,3,7,9\} & & N_{16} & \{7,9,17,19\} \\
N_4 & \{2,4,8,10\} & & N_{17} & \{10,12,18,20\} \\
N_5 & \{2,4,14,15\} & & N_{18} & \{10,12,24,26\} \\
N_6 & \{3,5,9,11\} & & N_{19} & \{11,12,19,20\} \\
N_7 & \{3,5,13,16\} & & N_{20} & \{11,12,25,26\} \\
N_8 & \{3,5,23,25\} & & N_{21} & \{13,14,15,16\} \\
N_9 & \{3,9,19,23\} & & N_{22} & \{13,16,23,25\} \\
N_{10} & \{4,6,10,12\} & & N_{23} & \{15,16,25,26\} \\
N_{11} & \{4,10,15,24\} & & N_{24} & \{17,18,19,20\} \\
N_{12} & \{5,6,11,12\} & & N_{25} & \{18,20,24,26\} \\
N_{13} & \{5,6,15,16\} & & N_{26} & \{19,20,25,26\}
\end{array} \tag{B.6}
$$

Each set in (B.6) denotes the corresponding vertices r_i. The remaining singular faces are

$$
\begin{array}{c|c|c}
D_{6,1} & \{r_1,r_2,r_3,r_4,r_5,r_6\} & r_{27}=(-1,-1,0,0) \\
D_{6,2} & \{r_7,r_8,r_9,r_{10},r_{11},r_{12}\} & r_{28}=(-1,0,0,0) \\
D_{7,1} & \{r_1,r_7,r_{13},r_{17},r_{21}\} & r_{29}=(0,0,-1,-1) \\
D_{7,2} & \{r_2,r_8,r_{14},r_{18},r_{22}\} & r_{30}=(0,0,-1,0),
\end{array} \tag{B.7}
$$

where the last column is the unique interior lattice point. These face data give precisely the singularity inventory in (5.17).

For definiteness, one smooth projective crepant subdivision is fixed as follows. On the thirty boundary rays assign heights

$$
b_i = 2^{26-i} \quad (1 \le i \le 26), \qquad b_i = -2^{30} + 2^{30-i} \quad (27 \le i \le 30). \tag{B.8}
$$

Take the lower-facet triangulation on every facet and cone it to the origin. The induced triangulations agree on common faces, every maximal cone is unimodular, and the four divisorial faces are star-triangulated at $r_{27},\ldots,r_{30}$. This fixes the exceptional curves and surfaces used below.

Principal-divisor relations eliminate the rays $15, 22, 24, 26$. Thus the rank-26 divisor basis is indexed by

$$
\mathcal{I} = \{1,2,\ldots,14,16,17,18,19,20,21,23,25,27,28,29,30\}. \tag{B.9}
$$

For a local coordinate a, write its charge row as $b_a = \sum_{I\in\mathcal{I}} B_{aI} D_I$. The complete sparse presentation of B is

$$
\begin{aligned}
b_{N_1} &= D_1 - D_2 - D_7 + D_8, & b_{N_2} &= D_1 - D_2 - D_{13} + D_{14},\\
b_{N_3} &= D_1 - D_3 - D_7 + D_9, & b_{N_4} &= D_2 - D_4 - D_8 + D_{10},\\
b_{N_5} &= D_2 - D_4 - D_{14}, & b_{N_6} &= D_3 - D_5 - D_9 + D_{11},\\
b_{N_7} &= D_3 - D_5 - D_{13} + D_{16}, & b_{N_8} &= D_3 - D_5 - D_{23} + D_{25},\\
b_{N_9} &= D_3 - D_9 + D_{19} - D_{23}, & b_{N_{10}} &= D_4 - D_6 - D_{10} + D_{12},\\
b_{N_{11}} &= D_4 - D_{10}, & b_{N_{12}} &= D_5 - D_6 - D_{11} + D_{12},\\
b_{N_{13}} &= D_5 - D_6 - D_{16}, & b_{N_{14}} &= D_5 - D_6 - D_{25},\\
b_{N_{15}} &= D_7 - D_8 - D_{17} + D_{18}, & b_{N_{16}} &= D_7 - D_9 - D_{17} + D_{19},\\
b_{N_{17}} &= D_{10} - D_{12} - D_{18} + D_{20}, & b_{N_{18}} &= D_{10} - D_{12},\\
b_{N_{19}} &= D_{11} - D_{12} - D_{19} + D_{20}, & b_{N_{20}} &= D_{11} - D_{12} - D_{25},\\
b_{N_{21}} &= D_{13} - D_{14} - D_{16}, & b_{N_{22}} &= D_{13} - D_{16} - D_{23} + D_{25},\\
b_{N_{23}} &= -D_{16} + D_{25}, & b_{N_{24}} &= D_{17} - D_{18} - D_{19} + D_{20},\\
b_{N_{25}} &= D_{18} - D_{20}, & b_{N_{26}} &= D_{19} - D_{20} - D_{25},\\
b_{D_{6,1},s_1} &= D_1 - D_2 - D_5 + D_6, & b_{D_{6,1},s_2} &= D_2 - D_3 - D_4 + D_5,\\
b_{D_{6,1},s_3} &= D_1 - D_2 - D_3 + D_4 + D_5 - D_6, & b_{D_{6,2},s_1} &= D_7 - D_8 - D_{11} + D_{12},\\
b_{D_{6,2},s_2} &= D_8 - D_9 - D_{10} + D_{11}, & b_{D_{6,2},s_3} &= D_7 - D_8 - D_9 + D_{10} + D_{11} - D_{12},\\
b_{D_{7,1},\alpha} &= -2D_1 + 2D_7 + D_{13} - D_{17}, & b_{D_{7,1},\beta} &= -D_1 + D_{13} + D_{17} - D_{21},\\
b_{D_{7,2},\alpha} &= 2D_2 - D_8 - 2D_{14}, & b_{D_{7,2},\beta} &= D_8 - D_{14} - D_{18}.
\end{aligned}
$$

This is a 36×26 matrix of rank 21, with twenty-one nonzero Smith invariants, all equal to one. Deleting the coordinates absent on a chosen E_3 component gives the four profile matrices in Table 3.

B.3 Primitive vectors in X°

If B denotes the 36×26 local-coordinate-by-divisor matrix, the five primitive vectors producing (5.19) are

$$
\begin{aligned}
v_9 &= \sum_{i=1}^{12} D_i + D_{17} + D_{18} + D_{19} + D_{20},\\
v_{11} &= -\sum_{i=7}^{12} D_i - D_{17} - D_{18} - D_{19} - D_{20} - D_{21},\\
v_{\beta_1} &= -D_{21},\\
v_{\beta_2} &= \sum_{i=1}^{12} D_i - D_{21},\\
v_{A_1} &= D_1 + D_3 + D_5 + D_7 + D_9 + D_{11} + D_{13} + D_{16} + D_{17} + D_{19} + D_{21} + D_{23} + D_{25}.
\end{aligned}
\tag{B.10}
$$

Direct multiplication gives

$$
\begin{aligned}
Bv_9 &= e_{u_9} - e_{\alpha_1} + e_{\alpha_2}, & Bv_{11} &= e_{u_{11}} - e_{\alpha_1} + e_{\alpha_2}, \\
Bv_{\beta_1} &= e_{\beta_1}, & Bv_{\beta_2} &= e_{\alpha_2} + e_{\beta_2}, \\
Bv_{A_1} &= e_{L_1} + e_{L_2}.
\end{aligned}
\tag{B.11}
$$

The complete sparse matrix is included in the reproducibility source described in Appendix C. Its Smith form has 21 nonzero unit invariants and five zero directions. After branch restriction, the four profiles have the ranks printed in Table 3.

C Reproducibility

All computations used in the physics translation are exact integer, rational or symbolic calculations. No numerical fit or floating-point rank decision enters a claim. From the project root, the local checks are run with

```
venv/bin/python checks/check_nodal_charge.py
venv/bin/python checks/check_e2_gauging.py
venv/bin/python checks/check_e3_gauging.py
venv/bin/python checks/check_e3_magnetic_quivers.py
venv/bin/python checks/check_xcirc_e3_profiles.py
venv/bin/python checks/check_x9_spectrum.py
python3 figures/code/compute04_smoothing_criterion.py
```

The scripts verify, respectively:

1. the nodal moment-map image and charge-kernel identity;
2. the E_2 invariant ring, balanced lift and X_9 reduced matrix;
3. the E_3 invariant ring, root basis and branchwise balanced lifts;
4. the affine-A_1/A_2 monopole sums and fiber-product Hilbert-series identity;
5. the four X° profile ranks, Smith invariants and isolating vectors;
6. the X_9 divisor quotient, Hodge numbers, effective vectors and multiplet changes; and
7. the three E_3 discriminant factors in the moment-map root convention, the finite row-space smoothing test, and the four profile restrictions of (5.22).

The source geometric computations are maintained with the companion mathematical papers. For the thirty-sector obstruction they include the independent reconstructions of the 36×26 matrix, its topological kernel, the local Milnor kernels and the vanishing-period statements. For X_9 they include the singular-locus calculation, branch restriction, explicit family, divisor quotient and Batyrev lattice-point counts [13, 23].

The finite scripts check the printed matrices and formulas. They do not replace the Hodge-theoretic proof of Theorem 4.2, the analytic necessity and sufficiency in Theorem 4.3, selected-base integrability, or the geometric construction of the X_9 family. The full toric input and deformation-base calculation for the thirty-node/two-cone counterexample, as well as the X_{19} smoothing, are reproduced in the companion mathematical source distribution [13].

The manuscript source and exact finite checks are available at `https://github.com/bwuebben/global-higgs-branch`.

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